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NATIONAL ASTRONOMICAL OBSERVATORIES, CAS

Pop II Top-heavy IMF Gate

背景、方法、结果与文献依据

中国科学院国家天文台

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目录

1. 背景：为什么需要 source-time gate
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3. 结果：Reed07 HMF + $z = 6$ calibration 后的 UVLF 响应
4. 文献依据：top-heavy / FFB / rapid growth 的外部支撑



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Part.01

背景

判定对象

- ▶ 判定单位是 Pop II 的 source-time SFR cell。
- ▶ Pop III 分支保持独立。
- ▶ Halo track 提供 $M_h(t')$ 、 $\dot{M}_h(t')$ 、 $z(t')$ 。

当前实现

- ▶ 对每个 active cell 计算 $I_{TH}(t')$ 。
- ▶ $I_{TH} = 1$: 该时刻 SFR 选 mild top-heavy Pop II SSP。
- ▶ $I_{TH} = 0$: 该时刻 SFR 选 canonical Pop II SSP。





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Part.02

方法

canonical component

top-heavy component

$$L_{\text{can}} = \int (1 - I_{\text{TH}}) \text{SFR}(t') \Psi_{\text{can}}(t_{\text{obs}} - t') dt' \quad L_{\text{TH}} = \int I_{\text{TH}} \text{SFR}(t') \Psi_{\text{TH}}(t_{\text{obs}} - t') dt'$$

- ▶ 总 UV 光度写成 $L_{\text{UV}} = L_{\text{can}} + L_{\text{TH}}$ 。
- ▶ MAH、SFR efficiency、dust mapping 保持一致；只切换 UV convolution 里的 SSP kernel。
- ▶ 因此它是一个受控的 SSP sensitivity test；自洽 variable-IMF galaxy model 另行处理。

Mode	判断条件	物理含义
canonical	$l_{\text{TH}}(\mathbf{t}') = 0$	永远不触发 top-heavy
z10_mild_topheavy	$l_{\text{TH}}(\mathbf{t}') = \mathbf{1}[z(\mathbf{t}') \geq z_{\text{TH}}]$	$z_{\text{TH}} = 10$ 以上 source time 全部切换
mah_burst_mild_topheavy	$\mathbf{1}[z(\mathbf{t}') \geq z_{\text{TH}}] \mathbf{1}[\tau_{\text{grow}}(\mathbf{t}') \leq \tau_{\text{th}}]$	只保留高红移且快速增长的子集

默认阈值

$$z_{\text{TH}} = 10, \quad \tau_{\text{th}} = 50 \text{ Myr}$$

MAH growth time

$$\tau_{\text{grow}}(\mathbf{t}') = \frac{M_{\text{h}}(\mathbf{t}')}{\dot{M}_{\text{h}}(\mathbf{t}')}$$

代码里只接受有限、正的 M_{h} 和 $\dot{M}_{\text{h}}$ 。

位置	关键对象	作用
auroralf/uvlf/imf.py auroralf/uvlf/pipeline.py	compute_topheavy_source_flags starforming_grid, topheavy_source_grid	$\text{mode} + z(t') + M_h/\dot{M}_h + \text{active mask} \rightarrow I_{\text{TH}}$ 。 只在正在形成星的 source-time cell 上判断 gate。
auroralf/uvlf/pipeline.py auroralf/uvlf/hmf_sampling.py	$L_{\text{can}}, L_{\text{TH}}$ sample weights	canonical 与 top-heavy SSP 分别卷积，再相加。 Reed07 HMF 权重进入 UVLF histogram。

一句话：top-heavy 的“哪部分”就是 `topheavy_source_grid=True` 的 source-time SFR；halo track 和 UVLF 只在之后聚合。

Step	条件	结果
1	SFR cell 不 active	canonical SSP
2	mode 是 canonical	canonical SSP
3	$z(t') < 10$	canonical SSP
4a	$z10_mild_topheavy$ 且 $z(t') \geq 10$	top-heavy SSP
4b	$mah_burst_mild_topheavy$ 且 $z(t') \geq 10$, $\tau_{grow} \leq 50$ Myr	top-heavy SSP
4c	$mah_burst_mild_topheavy$ 但 $\tau_{grow} > 50$ Myr	canonical SSP

最关键的一行

$$I_{TH}^{MAH}(t') = 1[z(t') \geq 10] \cdot 1\left[\frac{M_h(t')}{\dot{M}_h(t')} \leq 50 \text{ Myr}\right].$$



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Part.03

结果与结论

本版计算设置

- ▶ HMF: hmf package 的 Reed07 fitting function。
- ▶ redshift: $z = 6, 12.5, 14.5$ 。
- ▶ dust: 统一用同一套 $M_{UV}^{int} \rightarrow M_{UV}^{obs}$ mapping。

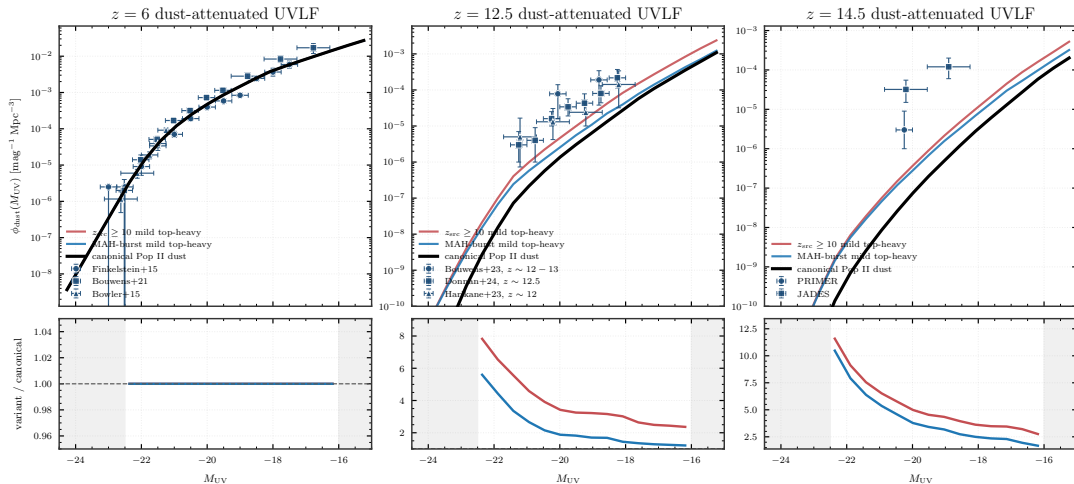
只在 $z = 6$ 调 ϵ_0

$$\epsilon_0 = 0.10212$$

由 $z = 6$ UVLF detections 的 log-space residual 标定；之后固定该值预测高红移。

为什么要重做

从历史 ST / massfunc HMF 换到 Reed07 后, $z = 6$ canonical UVLF 的归一化改变；不能沿用旧 ϵ_0 直接读 top-heavy 响应。



Redshift / range	$R_z = \phi_z / \phi_{\text{can}}$	$R_{\text{MAH}} = \phi_{\text{MAH}} / \phi_{\text{can}}$	解释
$z = 6$	1.00	1.00	calibration unchanged
$z = 12.5$, full	3.34	1.85	bright-end sensitive
$z = 12.5$, obs range	3.24	1.76	$-22.5 \leq M_{\text{UV}} \leq -16$
$z = 14.5$, obs range	4.44	3.30	higher rapid-growth fraction

结论

z-gate 给出上限响应；MAH gate 把 top-heavy 限制在 rapid-growth source-time 子集，因此更保守。

保持固定

- ▶ ϵ_0 只由 $z = 6$ 标定一次；高红移固定。
- ▶ SN feedback / gas recycling 固定。
- ▶ metal enrichment 与 dust production 固定。
- ▶ inferred stellar mass comparison 固定。

应该怎么表述

canonical Pop II 是 fiducial；两个 mild top-heavy mode 是文献动机下的 sensitivity mapping。

表达边界

避免把 MAH gate 说成自洽的气体密度、金属丰度或局域 SFH model；它只是 source-time proxy。

1. Top-heavy 的对象是 Pop II source-time SFR cell; halo track 与 Pop III 保持独立。
2. 判断条件只有两层: $z_{\text{src}} \geq 10$, 以及可选的 $\tau_{\text{grow}} = M_{\text{h}}/\dot{M}_{\text{h}} \leq 50 \text{ Myr}$ 。
3. Reed07 HMF 下, $z = 6$ 标定 $\epsilon_0 = 0.10212$; $z = 12.5$ 观测相关区间中, z-gate 约 3.24, MAH gate 约 1.76。



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Part.04

文献依据

Results. For this Evolving IMF model, we find that (i) the maximum UV luminosity enhancement is twice as large in massive galaxies ($\Delta M_{\text{UV}} \simeq 2.6$) than those where star formation is strongly limited by SN feedback ($\Delta M_{\text{UV}} \simeq 1.3$); (ii) it successfully reproduces the observed UV LFs at $z = 5 - 15$; (iii) galaxies with top-heavy IMFs exhibit the highest star formation rates, driven by their location in local density peaks, which facilitates higher gas accretion rates; (iv) the 1σ variances in the UV luminosity are only slightly higher compared to when assuming a Salpeter IMF, but the 2σ variances are significantly increased by a factor of $1.4 - 2$ boosting the abundance of UV-bright galaxies at $z > 10$; (v) reionisation begins earlier with more extended large ionised regions and fewer smaller ones during its initial stages, though these differences diminish at lower redshifts, leading to a similar end of reionisation at $z \simeq 5.6$.

Key words. Galaxies: high-redshift – Galaxies: evolution – Stars: mass function – intergalactic medium – dark ages, reionisation, first stars – Methods: numerical

Hutter et al. 2025, A&A / arXiv:2410.00730, p.1

UV 增亮

$\Delta M_{\text{UV}} \simeq 2.6$ in massive galaxies。

环境

highest SFR; local density peaks。

吸积

higher gas-accretion rates。

Galaxies with high sSFRs are typically gas-rich, sustained by high gas accretion rates that replenish their gas reservoirs and counteract the losses due to SNe. These galaxies are usually found in environments where they are the deepest potential wells or most massive galaxies (see e.g. Legrand et al. 2023). Consequently, low-mass galaxies with top-heavy IMFs are located in slightly above-average dense regions, while their massive counterparts are found in overdense regions, resulting in their median environmental density being slightly above that of the overall galaxy population. Furthermore, due to their higher gas accretion rates, which continuously dilute the ongoing metal enrichment, galaxies with top-heavy IMFs have the lowest gas-phase metallicities (oxygen-to-hydrogen mass ratio) despite their top-heavy IMFs producing in their SNe explosions more oxygen per

Hutter et al. 2025, Section 5, p.8

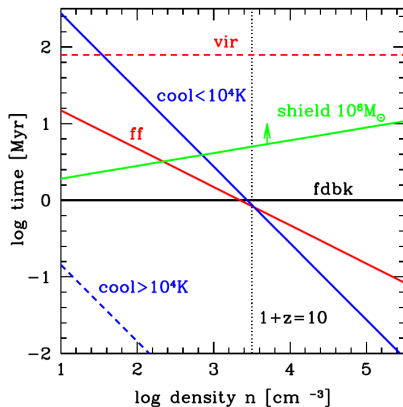
Gate: high gas accretion / rapid growth $\Rightarrow \tau_{\text{grow}}$ proxy

JWST observations indicate a surprising excess of luminous galaxies at $z \sim 10$ and above, consistent with efficient conversion of the accreted gas into stars, unlike the suppression of star formation by feedback at later times. We show that the high densities and low metallicities at this epoch *guarantee* a high star-formation efficiency (SFE) in the most massive dark-matter haloes. Feedback-free starbursts (FFBs) occur when the free-fall time is shorter than ~ 1 Myr, below the time for low-metallicity massive stars to develop winds and supernovae. This corresponds to a characteristic density of $\sim 3 \times 10^3 \text{ cm}^{-3}$. A comparable threshold density permits a starburst by allowing cooling to star-forming temperatures in a free-fall time. The galaxies within $\sim 10^{11} M_{\odot}$ haloes at $z \sim 10$ are expected to have FFB densities. The halo masses allow efficient gas supply by cold streams in a halo crossing time ~ 80 Myr. The FFBs gradually turn all the accreted gas into stars in clusters of $\sim 10^{4-7} M_{\odot}$ within galaxies that are rotating discs or shells. The starbursting clouds are insensitive to radiative feedback and are shielded against feedback from earlier stars. We predict high SFE above thresholds in redshift and halo mass, where the density is 10^{3-4} cm^{-3} . The $z \sim 10$ haloes of $\sim 10^{10.8} M_{\odot}$ are predicted to host galaxies of $\sim 10^{10} M_{\odot}$ with $\text{SFR} \sim 65 M_{\odot} \text{ yr}^{-1}$, blue colors, and sub-kpc sizes. The metallicity is $\leq 0.1 Z_{\odot}$ with little dust, gas, outflows and hot circum-galactic gas, allowing a top-heavy IMF but not requiring it. The compact galaxies with thousands of young FFB clusters may have implications on reionization, black-hole growth and globular clusters.

Key words: galaxies: evolution — galaxies: formation — galaxies: haloes — galaxies: interactions

[Dekel et al. 2023, MNRAS / arXiv:2303.04827, p.1](#)

抽取数字: $t_{\text{ff}} \lesssim 1 \text{ Myr}$, $n \sim 10^3\text{--}10^4 \text{ cm}^{-3}$, halo crossing $\sim 80 \text{ Myr}$, 并说明 top-heavy IMF is allowed but not required.



Dekel et al. 2023 给出的图像

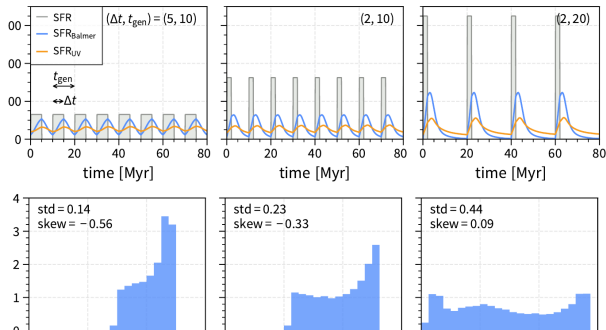
- ▶ feedback-free burst 需要 $t_{\text{ff}} \lesssim 1$ Myr。
- ▶ 对应 characteristic density 约 $10^3 - 10^4 \text{ cm}^{-3}$ 。
- ▶ halo crossing/gas supply 是更长的 ~ 80 Myr 量级。

Dekel et al. 2023, Fig. 3, p.5

The FFB phase of ~ 100 Myr is predicted
to show a characteristic SFH that fluctuates on a timescale of ~ 10 Myr.

Li et al. 2024, A&A / arXiv:2311.14662, p.1

取值含义: FFB phase ~ 100 Myr, characteristic SFH fluctuation ~ 10 Myr; 我们的 50 Myr 是 rapid-growth cut, 物理 burst duration 需要另行建模。



Li et al. 2024, Fig. 9, p.9

映射到 gate

- ▶ 图中 SFH 是 ~ 10 Myr fluctuation.
- ▶ 50 Myr 只筛 rapid-growth source time.
- ▶ 它比 ~ 80 – 100 Myr 供气/FFB 阶段更保守。

文献图像	放进模型的 proxy	对应 mode
$z > 10$ UV-bright tension highest SFR / gas-rich / high accretion	$z_{\text{src}} \geq 10$ $\tau_{\text{grow}} = M_{\text{h}}/\dot{M}_{\text{h}} \leq 50 \text{ Myr}$	z10_mild_topheavy mah_burst_mild_topheavy
FFB phase $\sim 80\text{--}100 \text{ Myr}$, bursty SFH $\sim 10 \text{ Myr}$	50 Myr rapid-growth cut; not burst duration	保守限制 bright-end 响应

讲的时候要强调

这是 literature-motivated source-time proxy; 它只标记 source-time 子集, 不把所有 $z > 10$ Pop II 统一设为 top-heavy, 也不把 MAH gate 等同于 gas density 或 metallicity model。

- ▶ Hutter et al. 2025, p.1: abstract results; top-heavy IMF, UV boost, density peaks, gas accretion
- ▶ Hutter et al. 2025, p.8: high-sSFR / gas-rich / high-accretion environment
- ▶ Dekel et al. 2023, p.1: FFB abstract; 1 Myr, $10^3\text{--}10^4\text{ cm}^{-3}$, 80 Myr
- ▶ Dekel et al. 2023, p.5: timescale-density Fig. 3
- ▶ Li et al. 2024, p.1: FFB phase ~ 100 Myr, SFH fluctuation ~ 10 Myr
- ▶ Li et al. 2024, p.9: bursty SFH Fig. 9

备注

这些链接是外部 PDF 页码锚点；大多数 PDF viewer / browser 会直接跳到对应页。

感谢各位老师指正